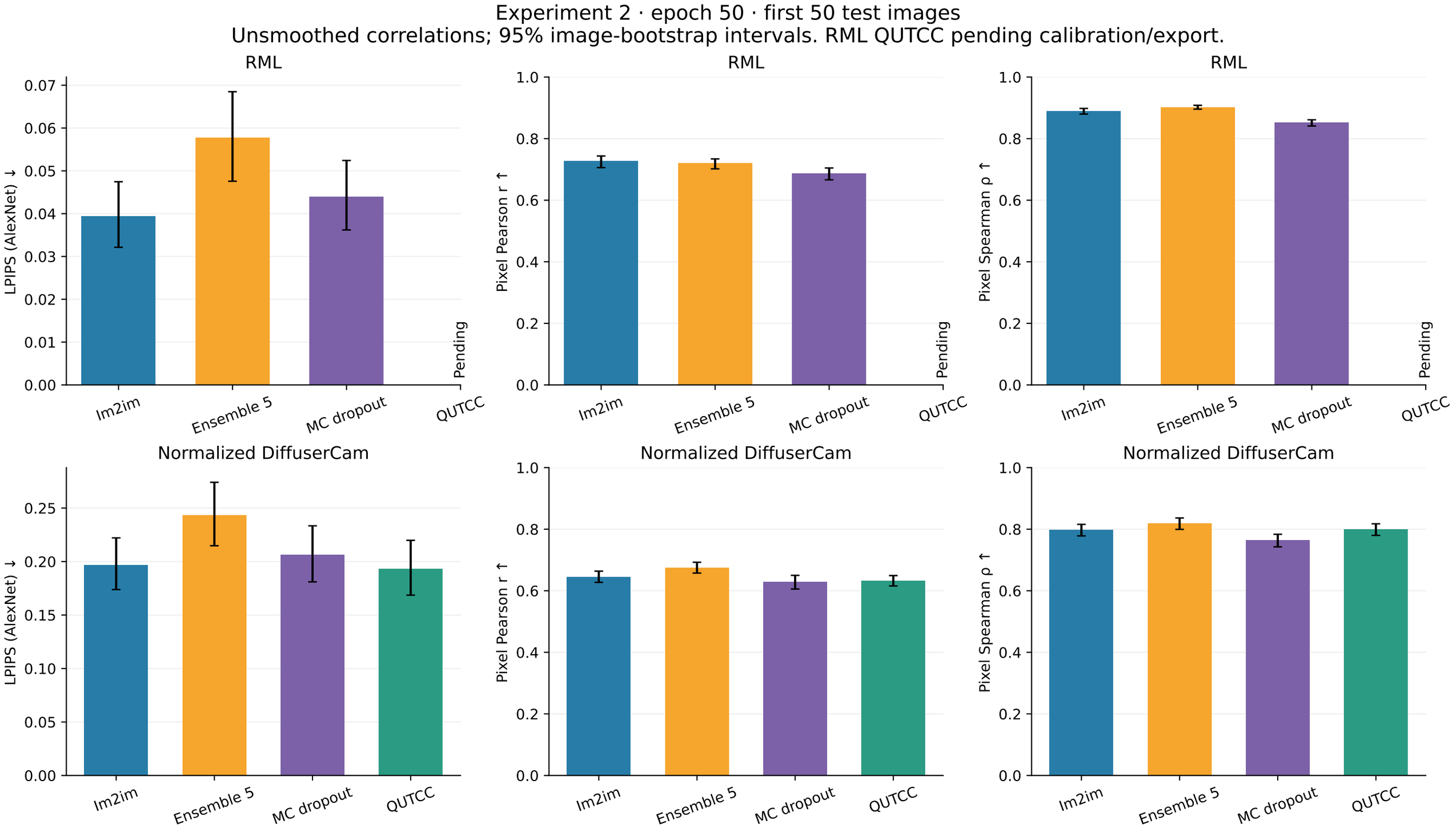
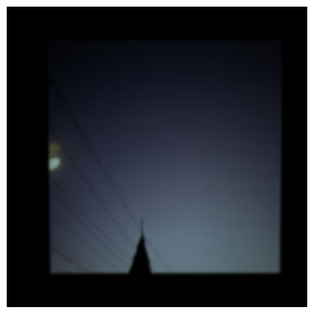

Experiment 2 · epoch 50 · first 50 test images
Unsmoothed correlations; 95% image-bootstrap intervals. RML QUTCC pending calibration/export.
RML
LPIPS (AlexNet) ↓
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
Im2im
Ensemble 5
MC dropout
QUTCC
Pending
RML
Pixel Pearson r ↑
0.0
0.2
0.4
0.6
0.8
1.0
Im2im
Ensemble 5
MC dropout
QUTCC
Pending
RML
Pixel Spearman ρ ↑
0.0
0.2
0.4
0.6
0.8
1.0
Im2im
Ensemble 5
MC dropout
QUTCC
Pending
Normalized DiffuserCam
LPIPS (AlexNet) ↓
0.00
0.05
0.10
0.15
0.20
0.25
Im2im
Ensemble 5
MC dropout
QUTCC
Normalized DiffuserCam
Pixel Pearson r ↑
0.0
0.2
0.4
0.6
0.8
1.0
Im2im
Ensemble 5
MC dropout
QUTCC
Normalized DiffuserCam
Pixel Spearman ρ ↑
0.0
0.2
0.4
0.6
0.8
1.0
Im2im
Ensemble 5
MC dropout
QUTCC

RML · test example 1 · ID 35894
All correlations use the full 256×256 map; scatter displays a fixed 4,096-pixel subset.

Measurement

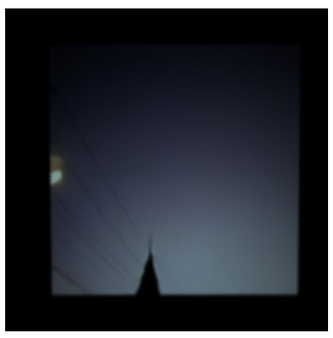


Ground truth

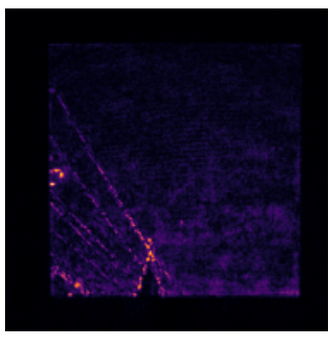


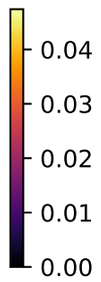
Error = mean RGB |prediction − truth|
Uncertainty = mean RGB (upper − lower)
Shared map scale across models and both quantities.
Display saturates at the 99.5th percentile;
correlations use original values, without smoothing.

Im2im
LPIPS 0.01472

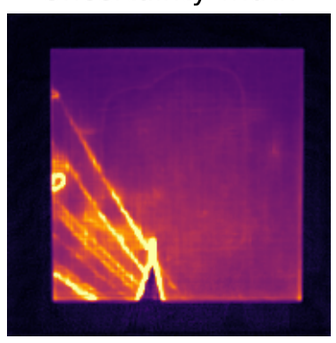


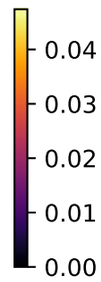
Absolute error



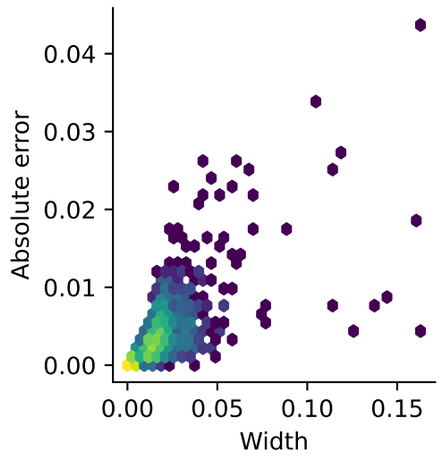


Uncertainty width

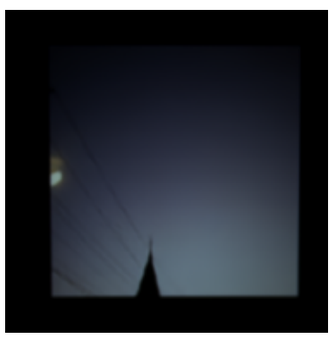




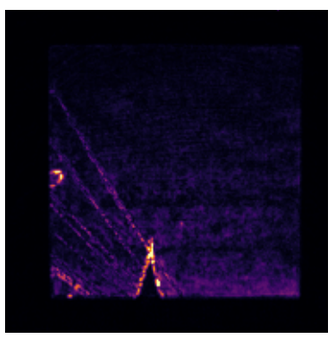
Pearson 0.75528
Spearman 0.87065

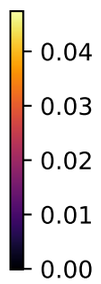


Ensemble (5)
LPIPS 0.02127

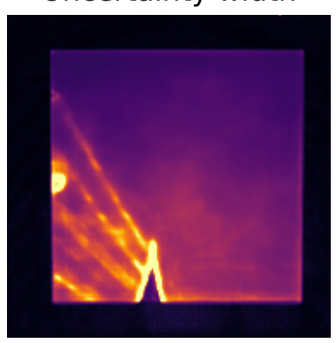


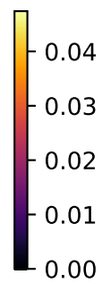
Absolute error



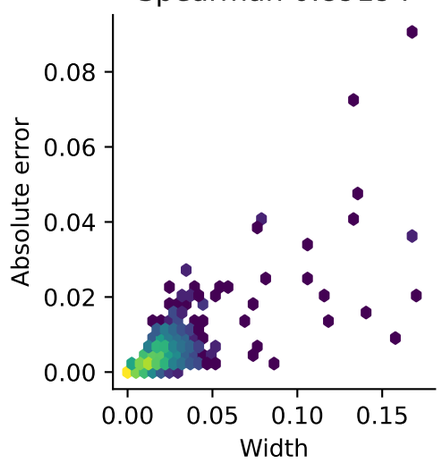


Uncertainty width





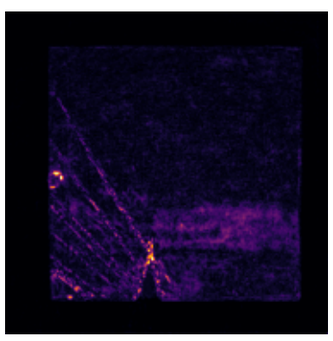
Pearson 0.78494
Spearman 0.89194

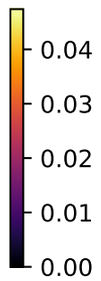


dropout (p=0.05, 30 passes)
LPIPS 0.01237

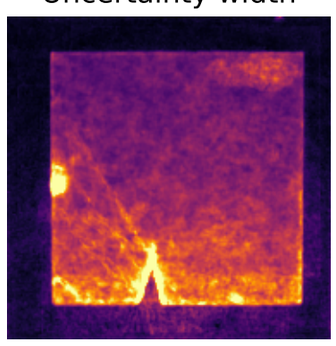


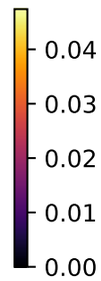
Absolute error



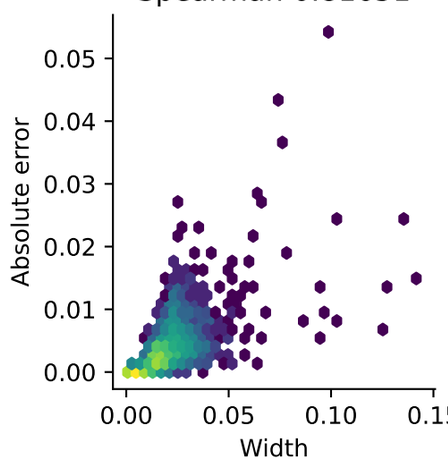


Uncertainty width





Pearson 0.69132
Spearman 0.81651



QUTCC epoch-50 result pending calibration/export

RML · test example 2 · ID 24193
All correlations use the full 256×256 map; scatter displays a fixed 4,096-pixel subset.

Measurement



Ground truth

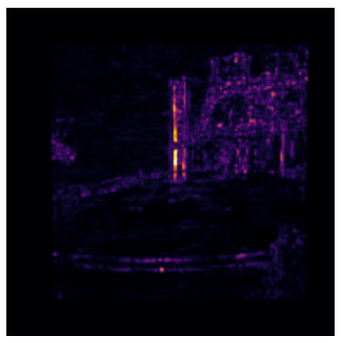


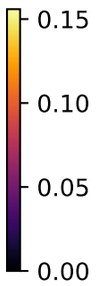
Error = mean RGB |prediction − truth|
Uncertainty = mean RGB (upper − lower)
Shared map scale across models and both quantities.
Display saturates at the 99.5th percentile;
correlations use original values, without smoothing.

Im2im
LPIP 0.04026

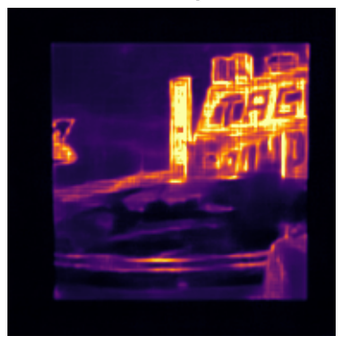


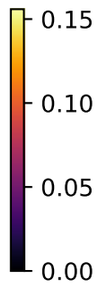
Absolute error

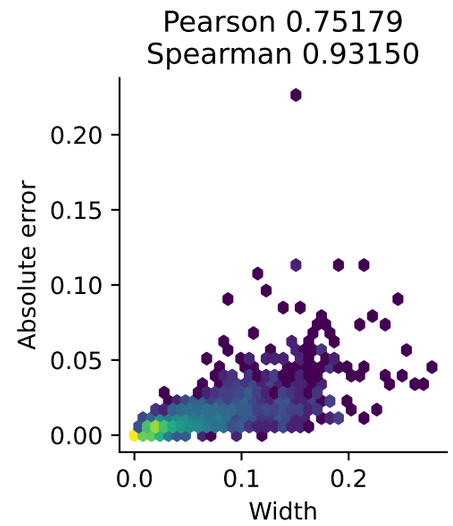




Uncertainty width



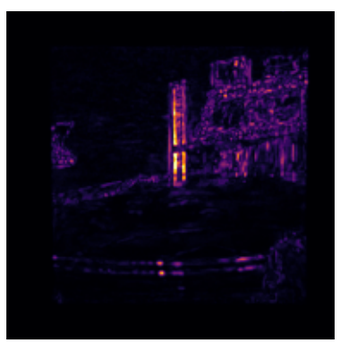


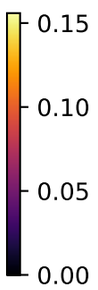


Ensemble (5)
LPIPS 0.06941

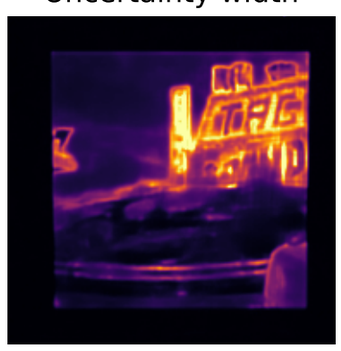


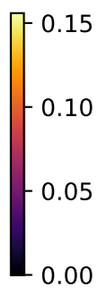
Absolute error

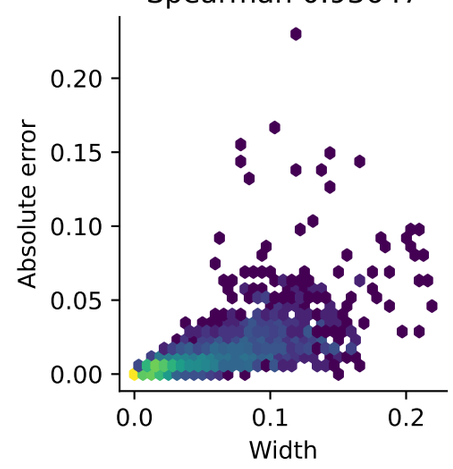




Uncertainty width



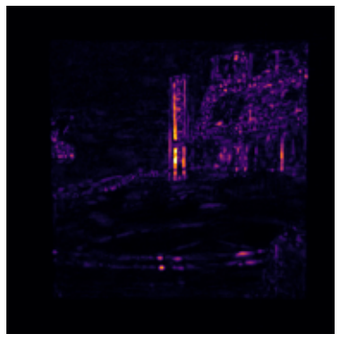


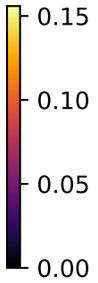


dropout (p=0.05, 30 passes)
LPIPS 0.04176

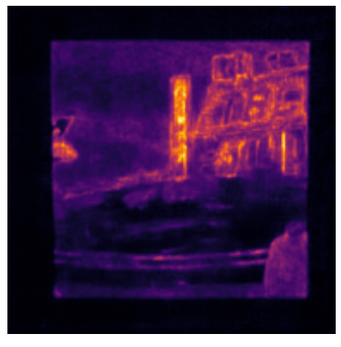


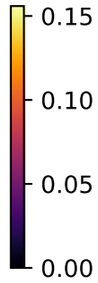
Absolute error

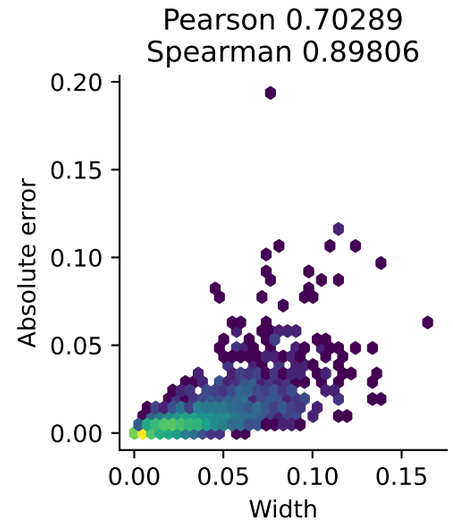




Uncertainty width







QUTCC epoch-50 result pending calibration/export

RML · test example 6 · ID 68127
All correlations use the full 256×256 map; scatter displays a fixed 4,096-pixel subset.

Measurement



Ground truth

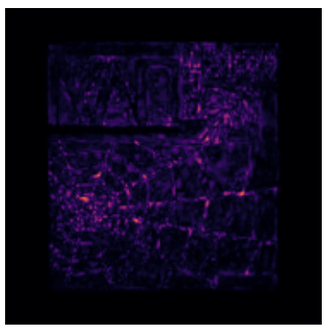


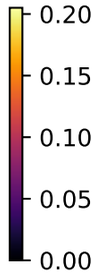
Error = mean RGB |prediction − truth|
Uncertainty = mean RGB (upper − lower)
Shared map scale across models and both quantities.
Display saturates at the 99.5th percentile;
correlations use original values, without smoothing.

Im2im
LPIPS 0.02093

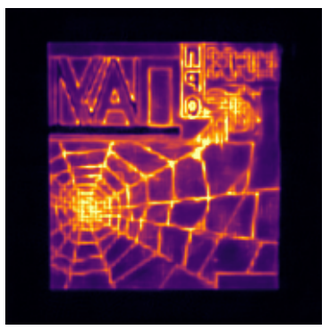


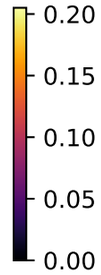
Absolute error

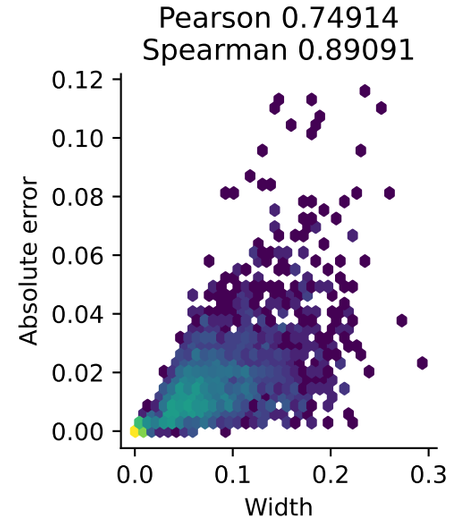




Uncertainty width



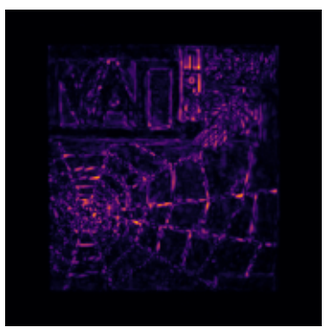


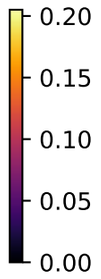


Ensemble (5)
LPIPS 0.04944

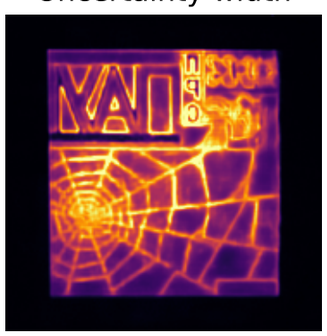


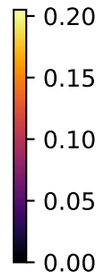
Absolute error

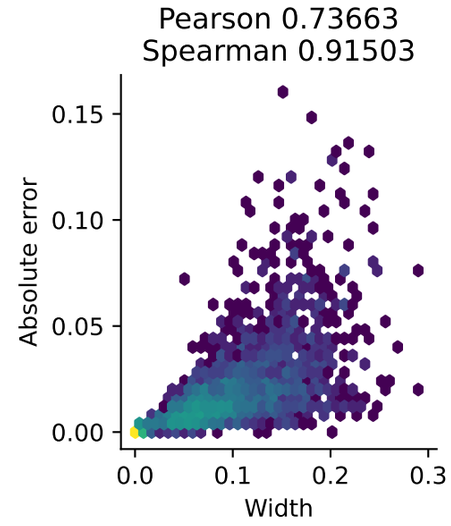




Uncertainty width



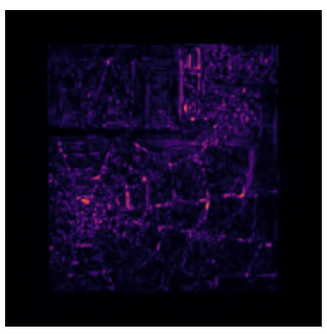


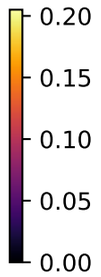


ropout (p=0.05, 30 passes)
LPIPS 0.02516

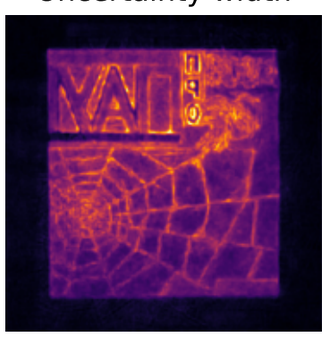


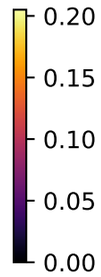
Absolute error

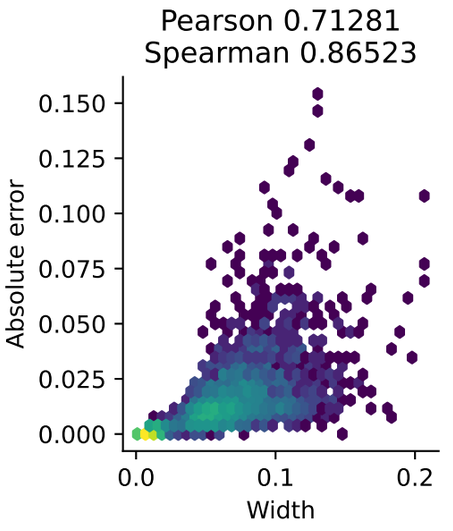




Uncertainty width



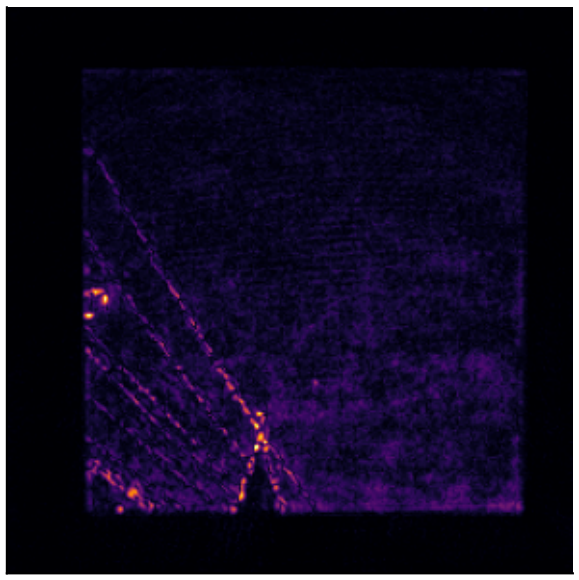


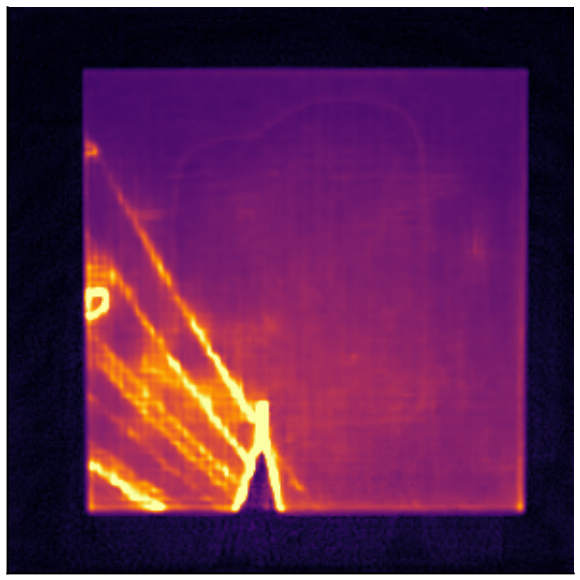


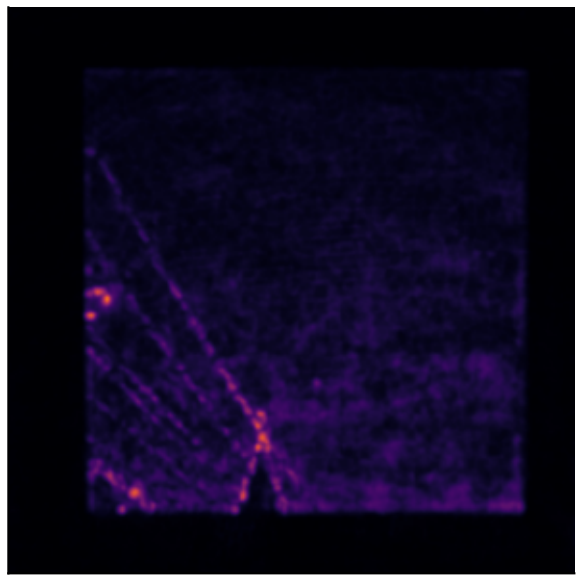
QUTCC epoch-50 result pending calibration/export

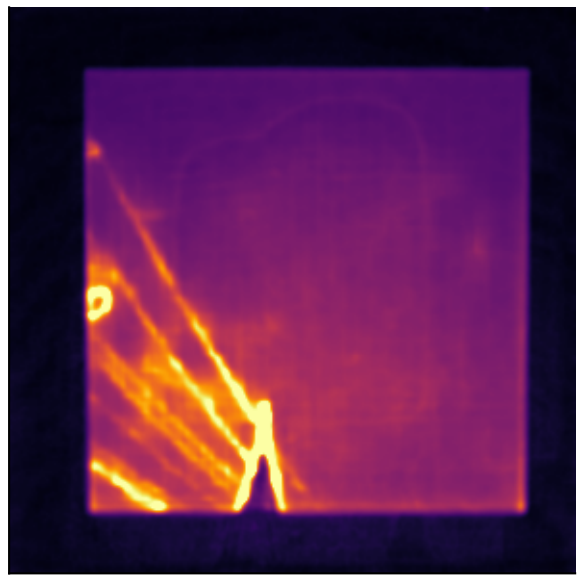
rml · example 1 · raw versus smoothed maps (same display scale)

Im2im

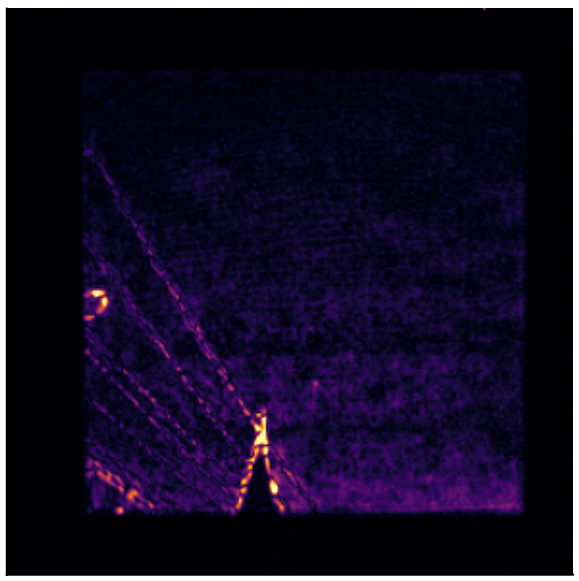


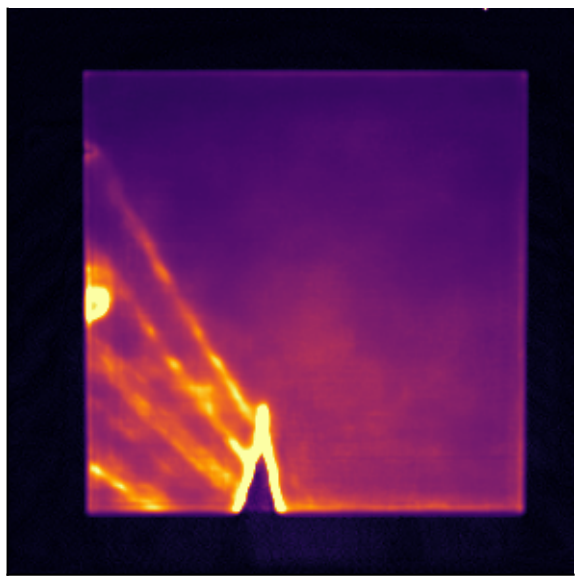


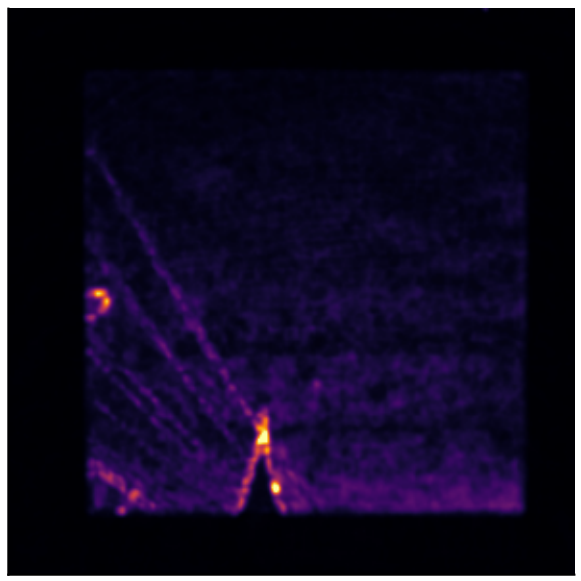


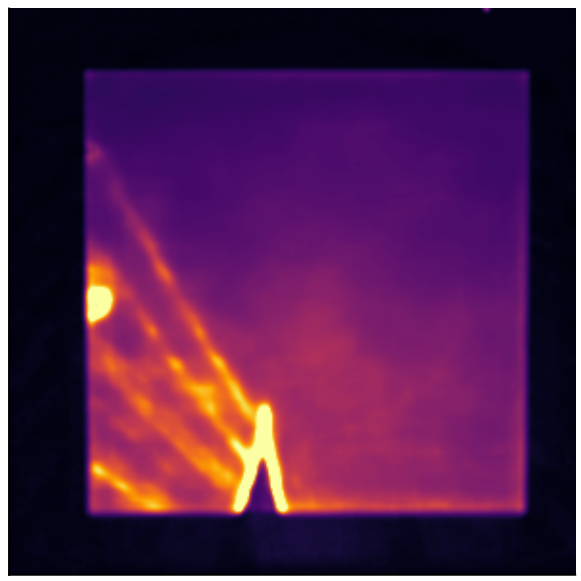


Ensemble (5)

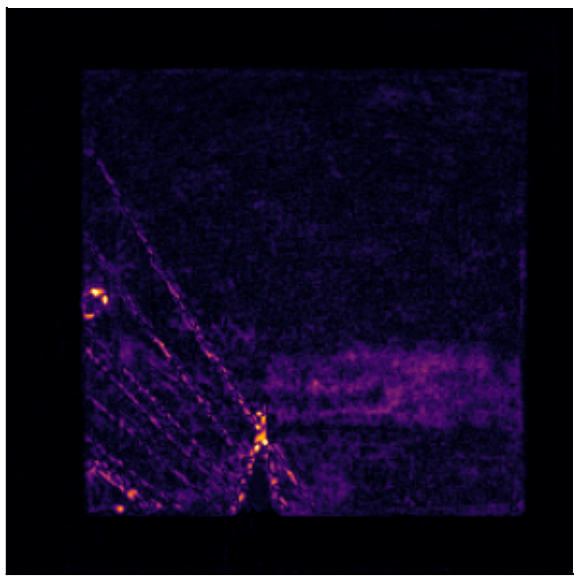


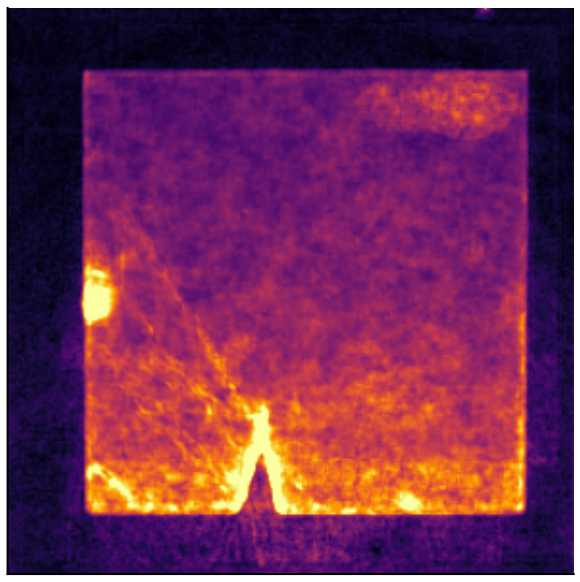


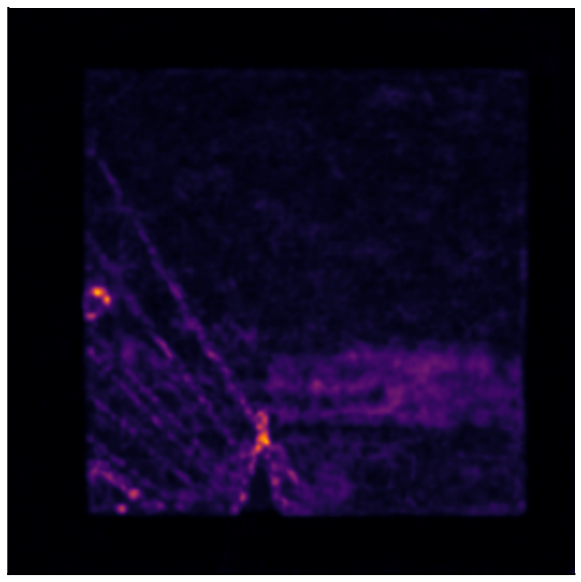


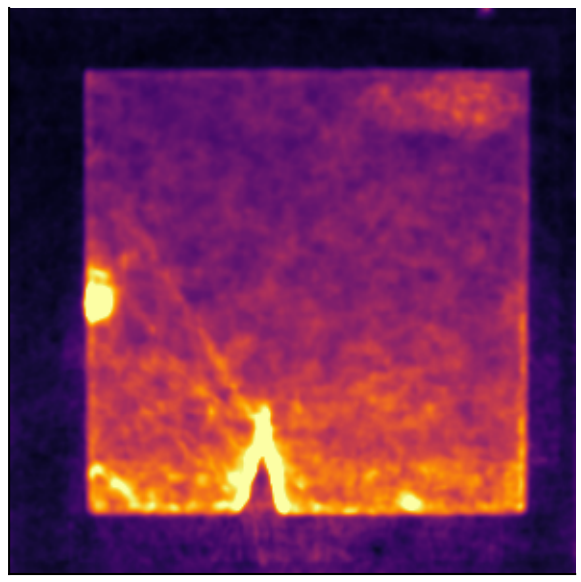


MC dropout (p=0.05, 30 passes)







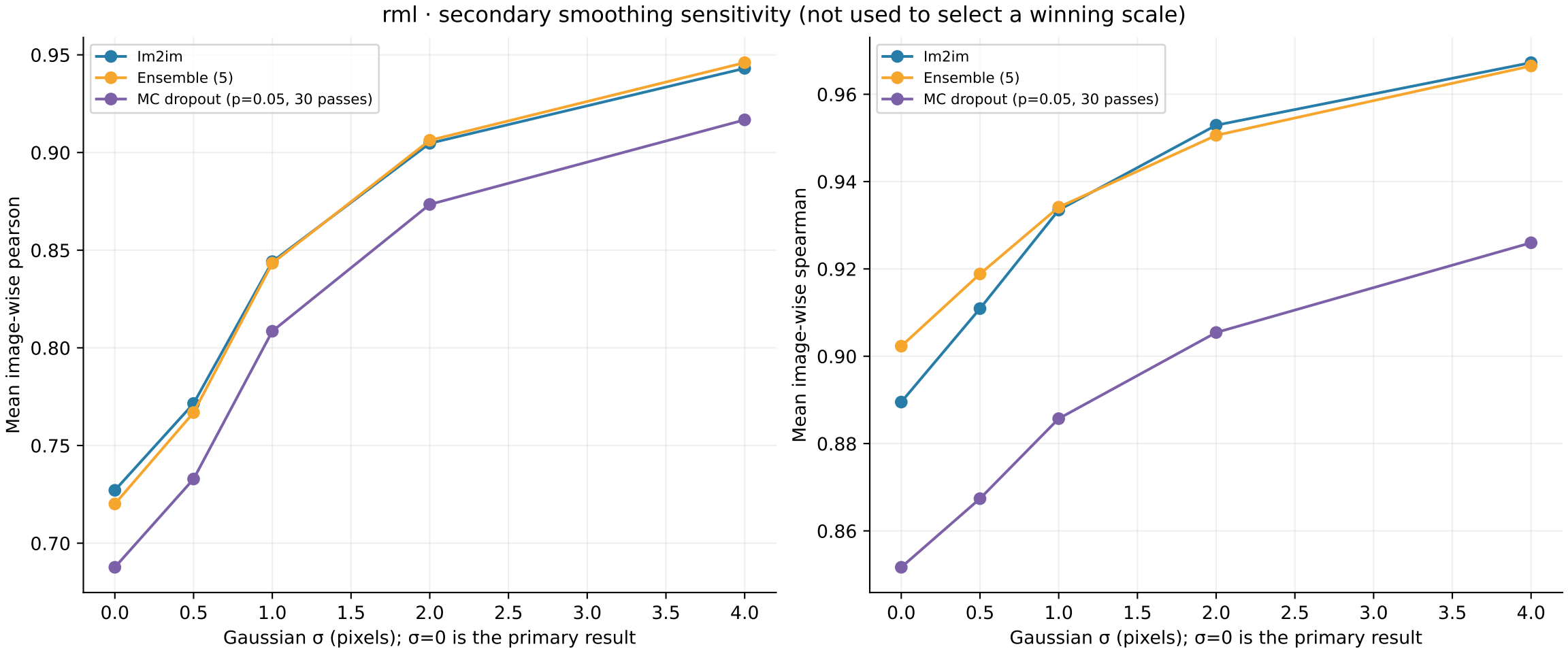


Pending

Pending

Pending

Pending


rml · secondary smoothing sensitivity (not used to select a winning scale)
Im2im
Ensemble (5)
MC dropout (p=0.05, 30 passes)
Mean image-wise pearson
0.95
0.90
0.85
0.80
0.75
0.70
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Gaussian σ (pixels); σ=0 is the primary result
Im2im
Ensemble (5)
MC dropout (p=0.05, 30 passes)
Mean image-wise spearman
0.96
0.94
0.92
0.90
0.88
0.86
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Gaussian σ (pixels); σ=0 is the primary result

Normalized DiffuserCam · test example 1 · ID 35894
All correlations use the full 256×256 map; scatter displays a fixed 4,096-pixel subset.

Measurement

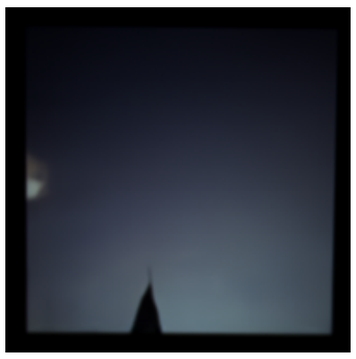


Ground truth

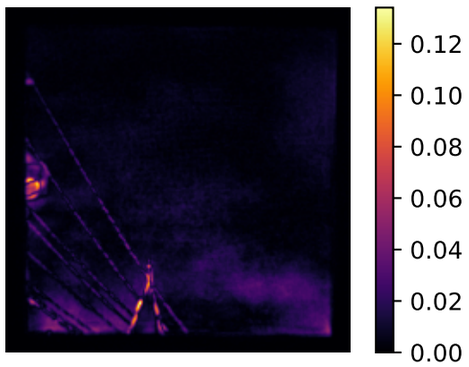


Error = mean RGB |prediction − truth|
Uncertainty = mean RGB (upper − lower)
Shared map scale across models and both quantities.
Display saturates at the 99.5th percentile;
correlations use original values, without smoothing.

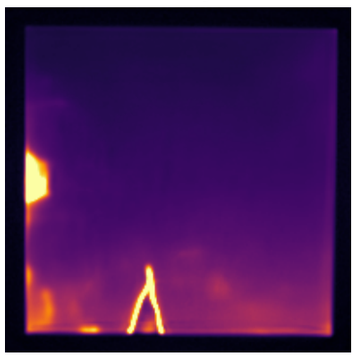
Im2im
LPIPS 0.07865



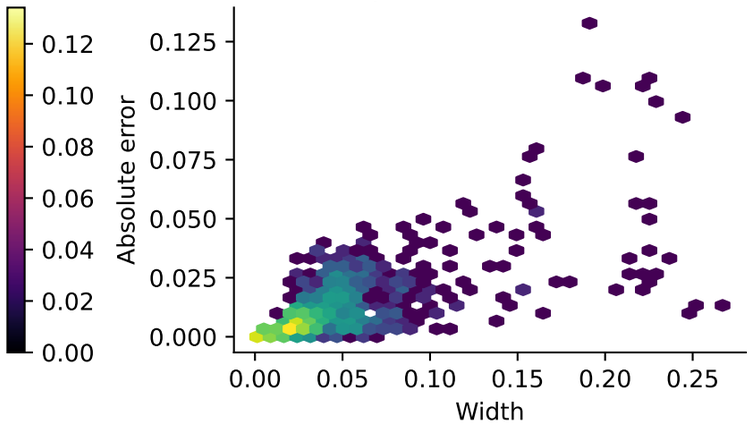
Absolute error



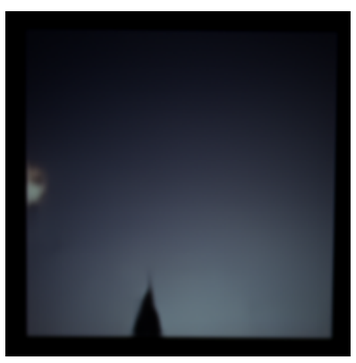
Uncertainty width



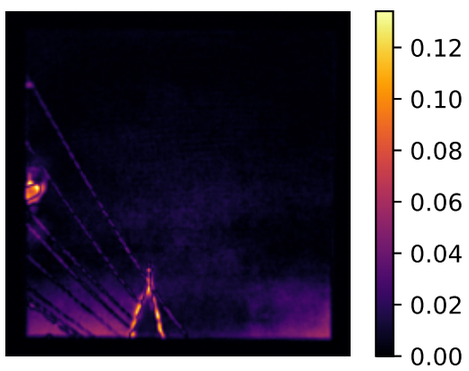
Pearson 0.72715
Spearman 0.78600



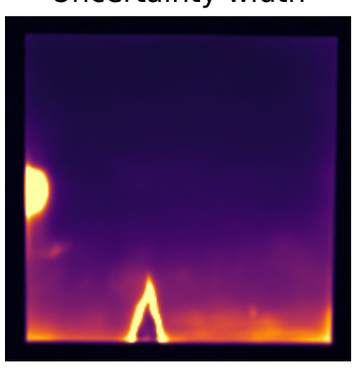
Ensemble (5)
LPIPS 0.08761



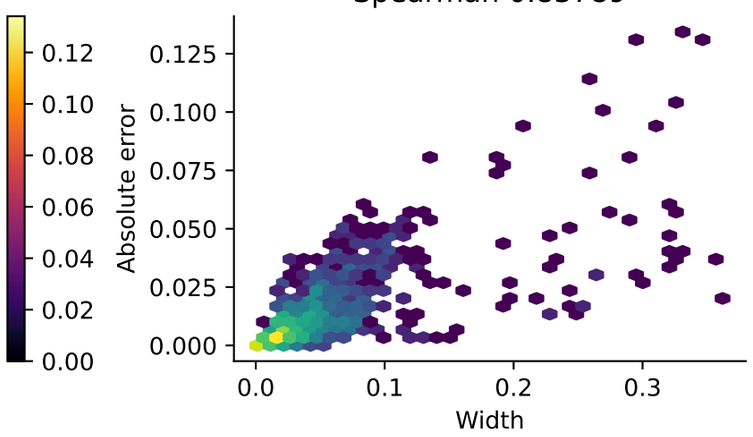
Absolute error



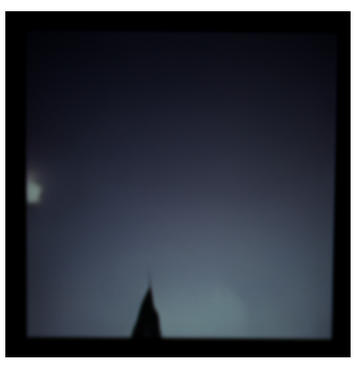
Uncertainty width



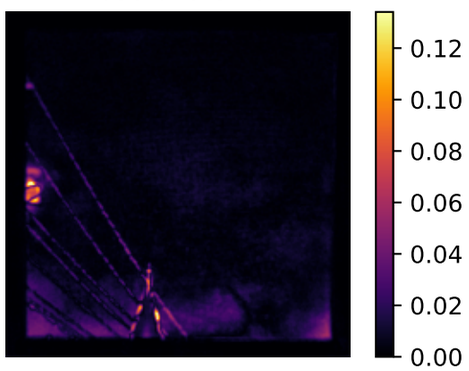
Pearson 0.75988
Spearman 0.83789



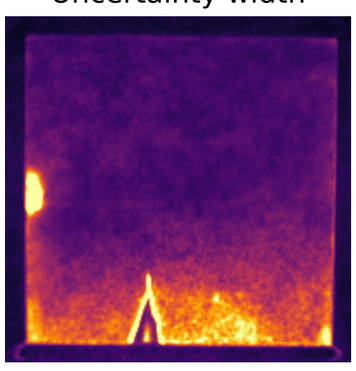
MC dropout (p=0.05, 30 passes)
LPIPS 0.07967



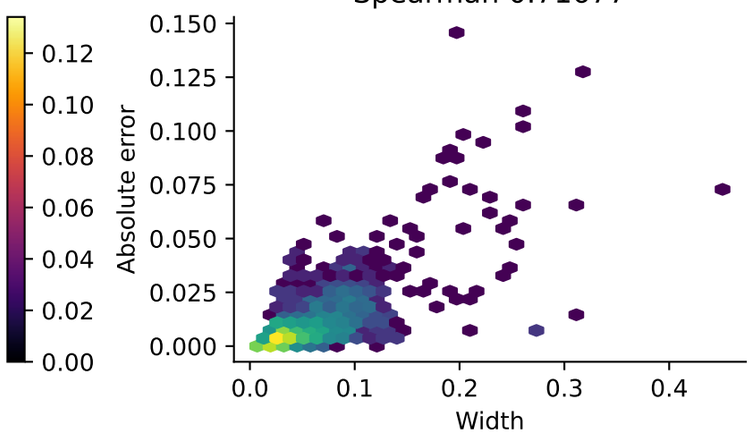
Absolute error



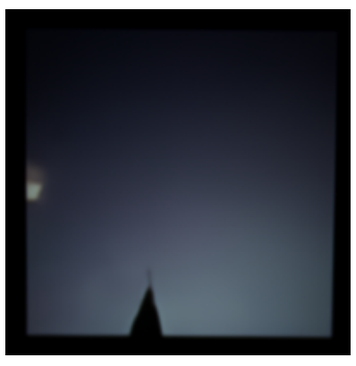
Uncertainty width



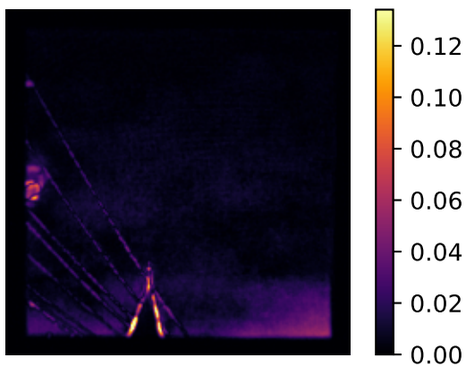
Pearson 0.71268
Spearman 0.71677



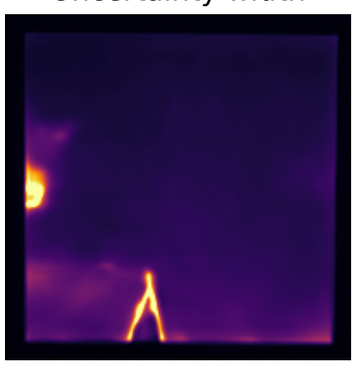
QUTCC
LPIPS 0.07983



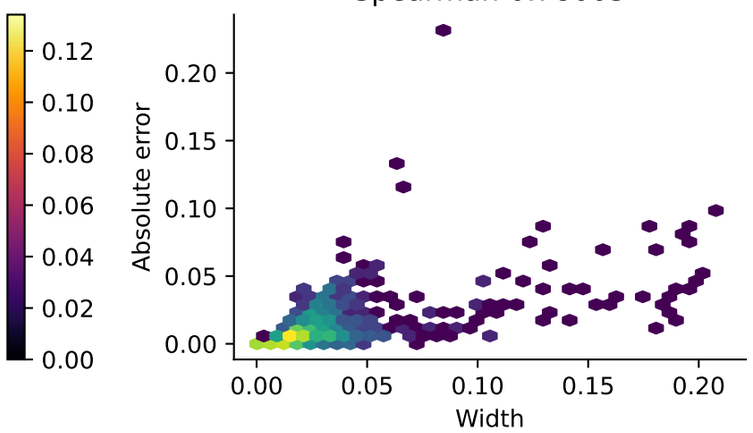
Absolute error



Uncertainty width



Pearson 0.64844
Spearman 0.78003



Normalized DiffuserCam · test example 2 · ID 24193
All correlations use the full 256×256 map; scatter displays a fixed 4,096-pixel subset.

Measurement



Ground truth

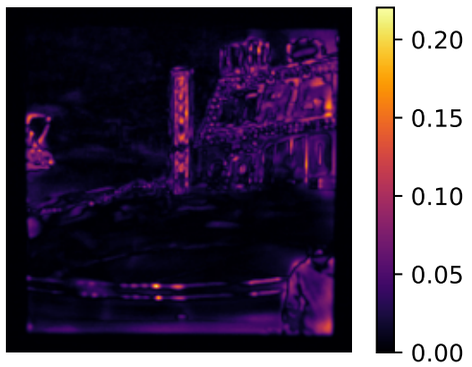


Error = mean RGB |prediction − truth|
Uncertainty = mean RGB (upper − lower)
Shared map scale across models and both quantities.
Display saturates at the 99.5th percentile;
correlations use original values, without smoothing.

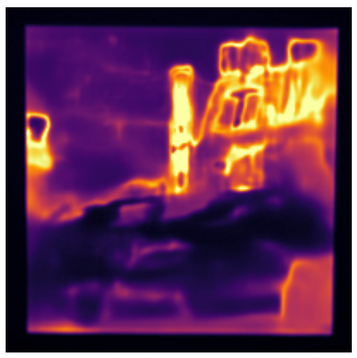
Im2im
LPIPS 0.27500



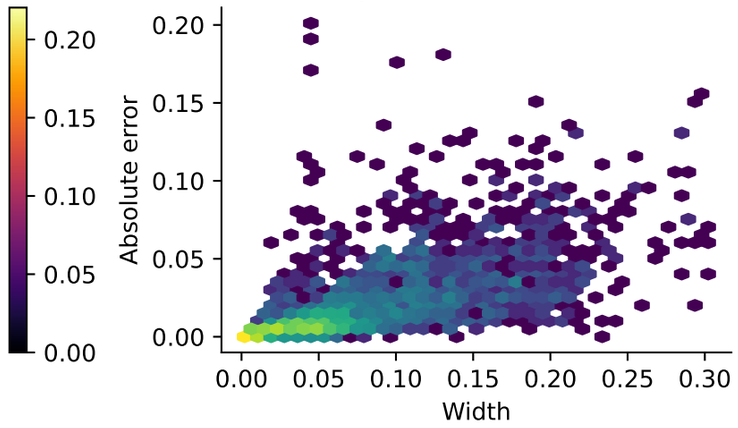
Absolute error



Uncertainty width



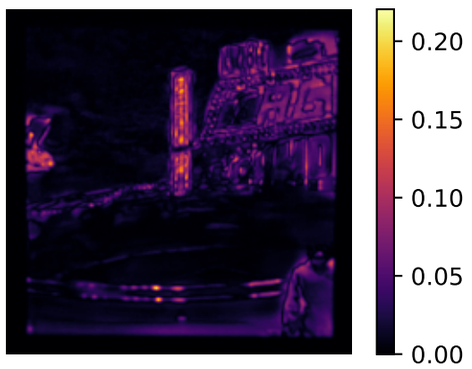
Pearson 0.72152
Spearman 0.84947



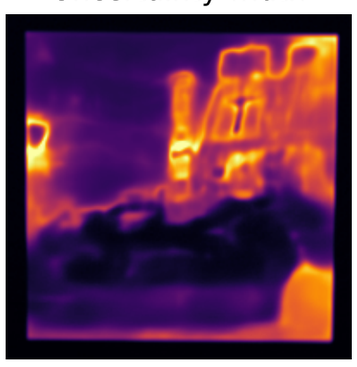
Ensemble (5)
LPIPS 0.34091



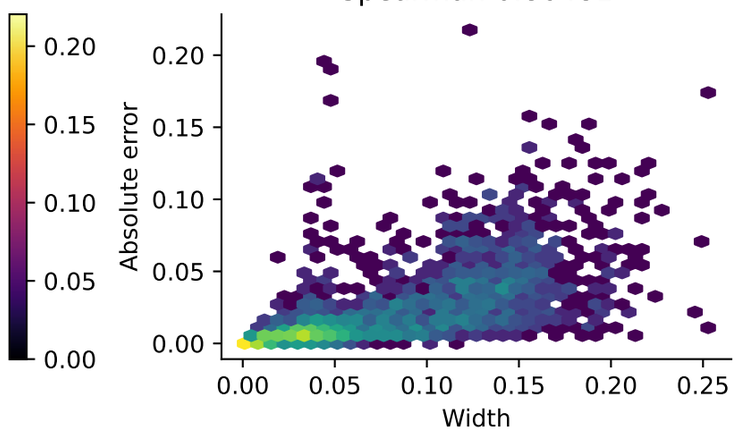
Absolute error



Uncertainty width



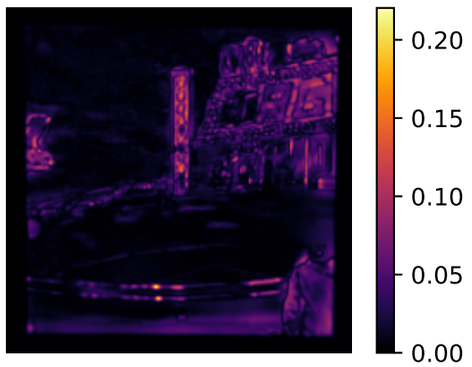
Pearson 0.72599
Spearman 0.86481



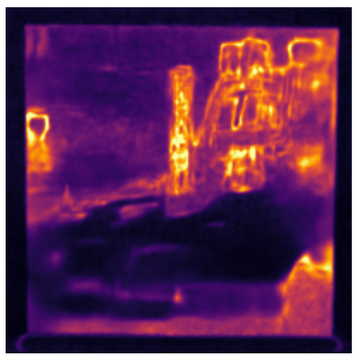
MC dropout (p=0.05, 30 passes)
LPIPS 0.28697



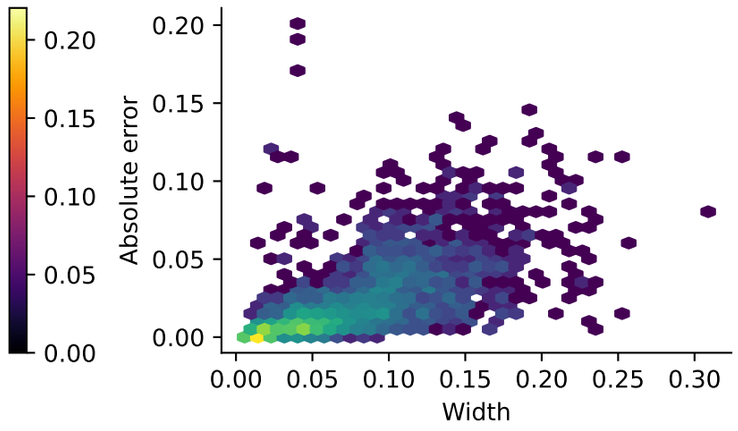
Absolute error



Uncertainty width



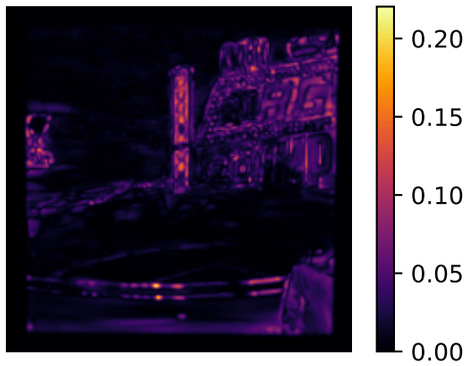
Pearson 0.72599
Spearman 0.83907



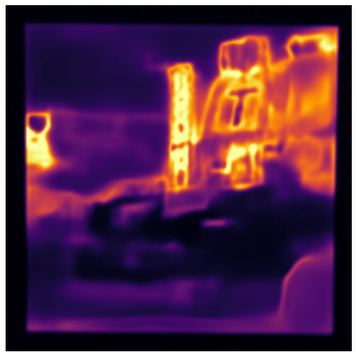
QUTCC
LPIPS 0.26939



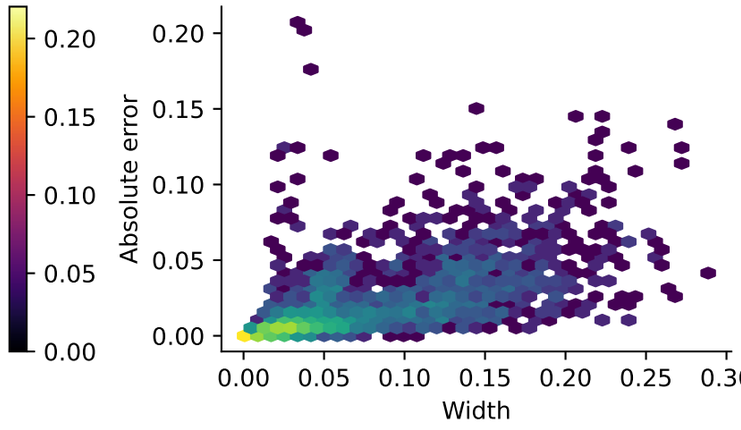
Absolute error



Uncertainty width



Pearson 0.70438
Spearman 0.85539



Normalized DiffuserCam · test example 6 · ID 68127
All correlations use the full 256×256 map; scatter displays a fixed 4,096-pixel subset.

Measurement



Ground truth

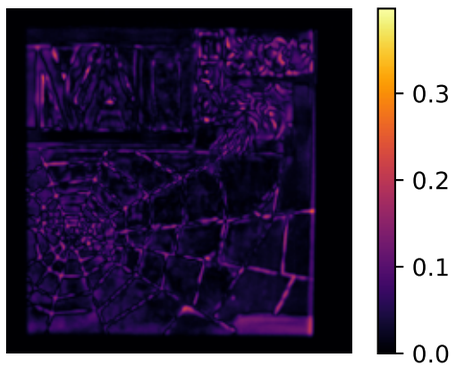


Error = mean RGB |prediction − truth|
Uncertainty = mean RGB (upper − lower)
Shared map scale across models and both quantities.
Display saturates at the 99.5th percentile;
correlations use original values, without smoothing.

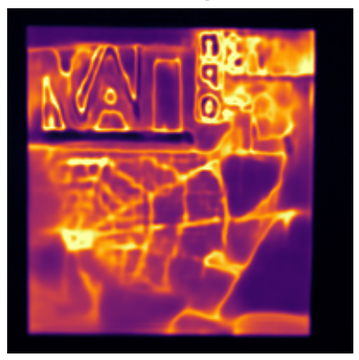
Im2im
LPIPS 0.22601

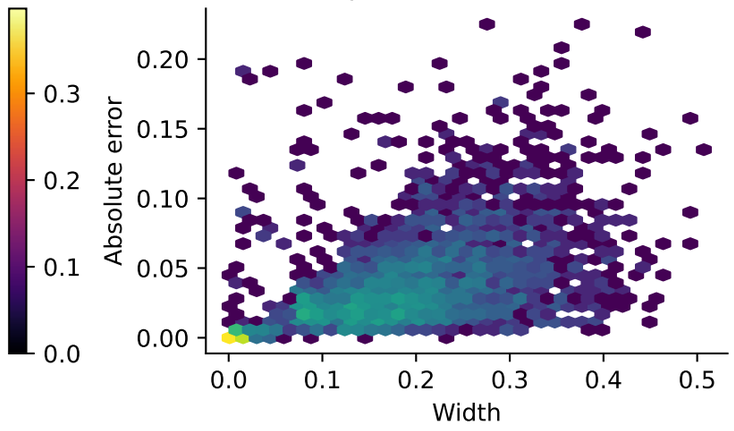


Absolute error



Uncertainty width



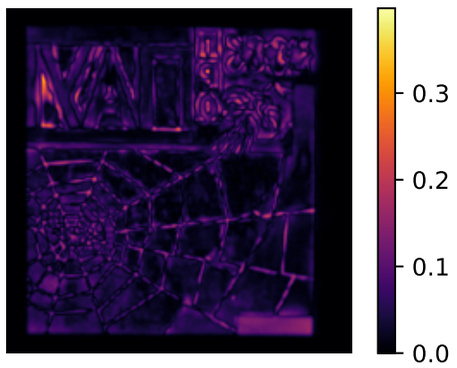


Pearson 0.61863
Spearman 0.75920

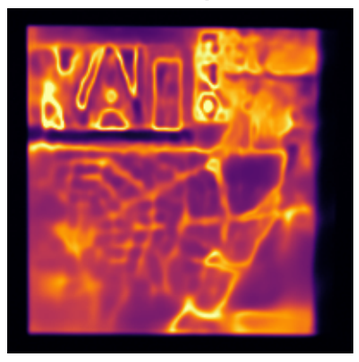
Ensemble (5)
LPIPS 0.35668

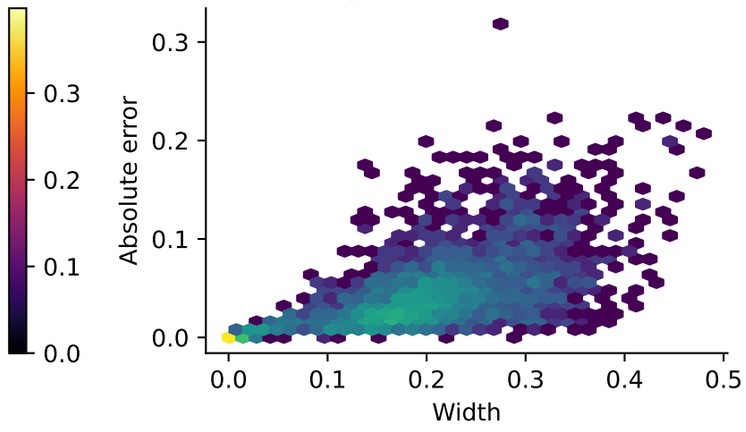


Absolute error



Uncertainty width



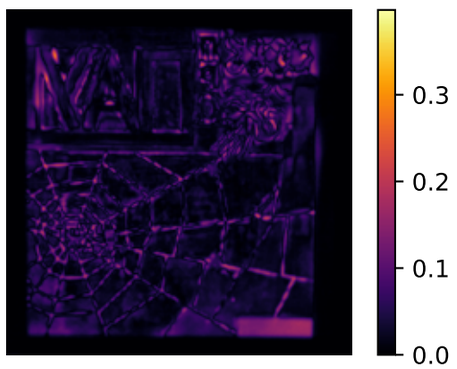


Pearson 0.70357
Spearman 0.81222

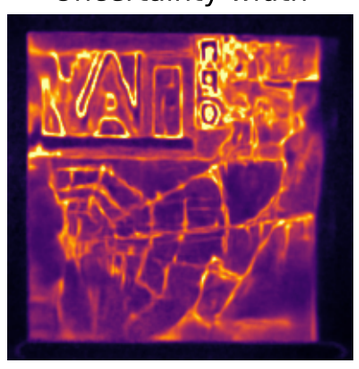
MC dropout (p=0.05, 30 passes)
LPIPS 0.24110

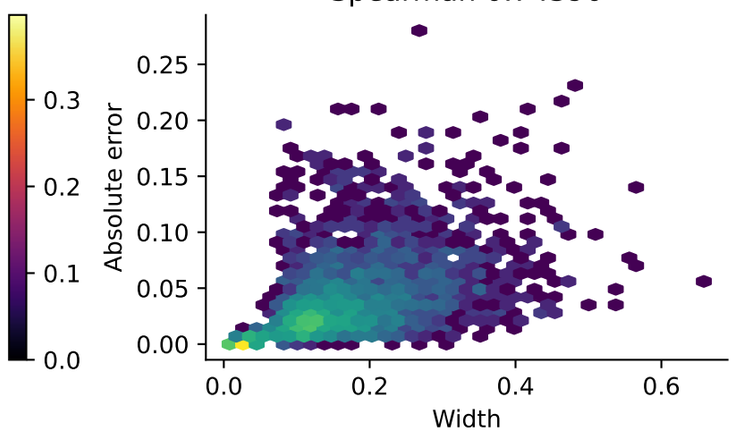


Absolute error



Uncertainty width



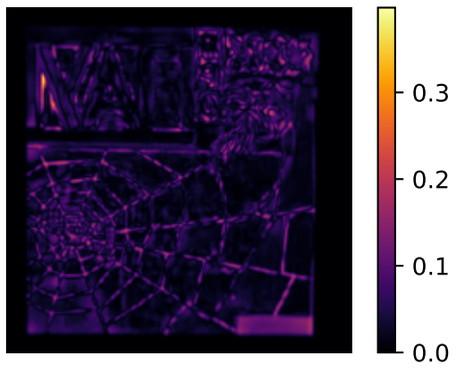


Pearson 0.57708
Spearman 0.74390

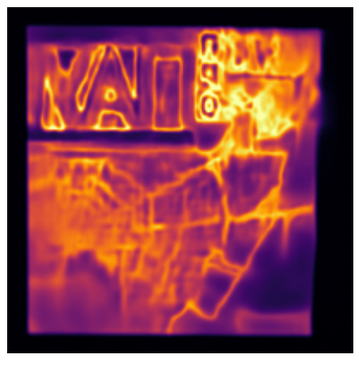
QUTCC
LPIPS 0.22445

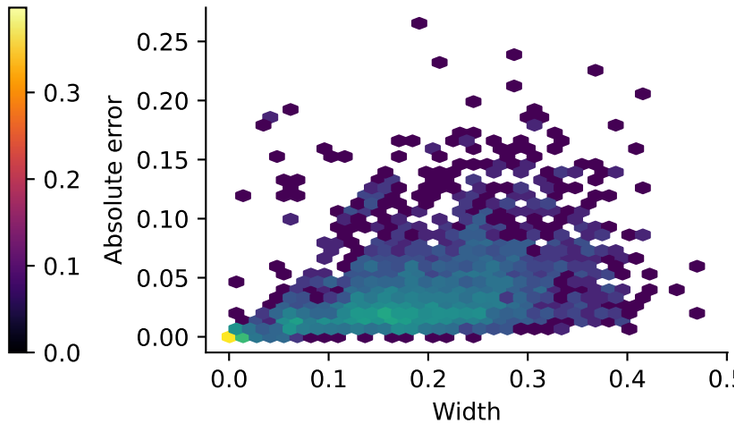


Absolute error



Uncertainty width

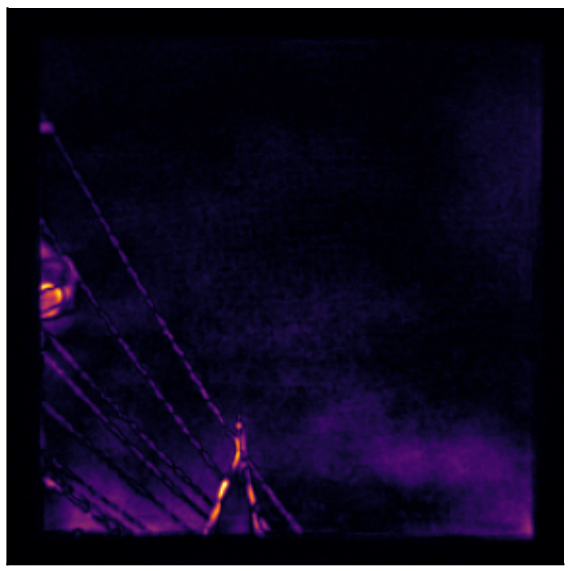


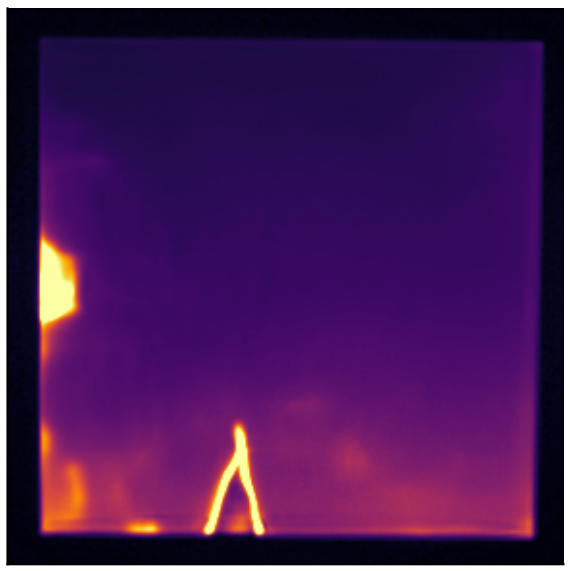


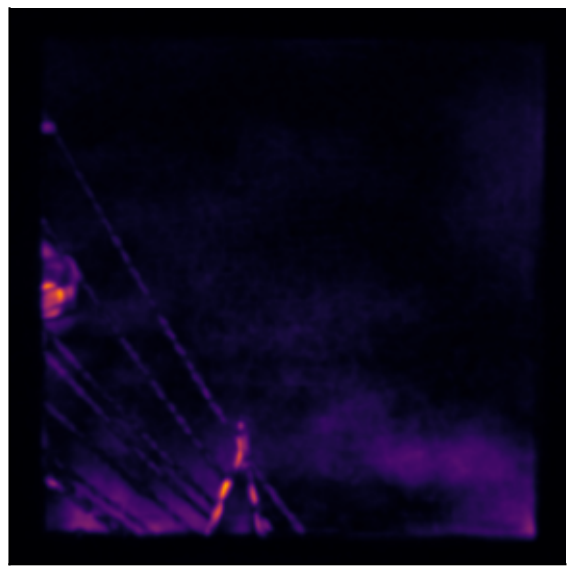
Pearson 0.57832
Spearman 0.74187

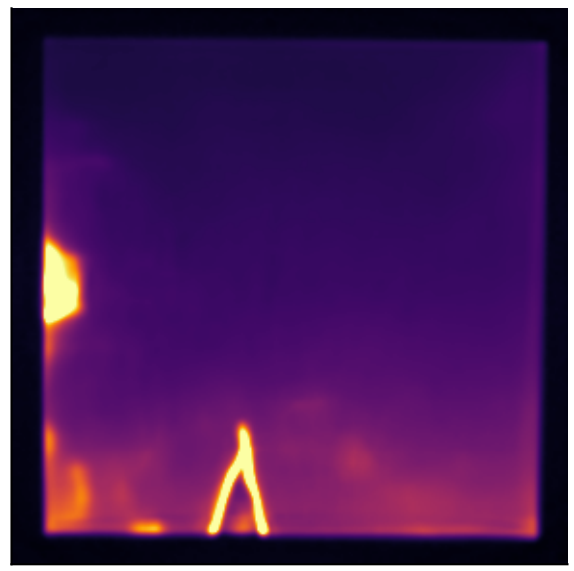
diffuser · example 1 · raw versus smoothed maps (same display scale)

Im2im

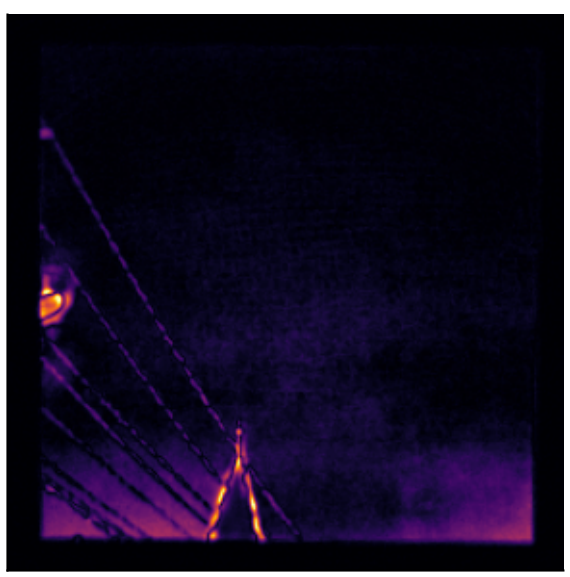


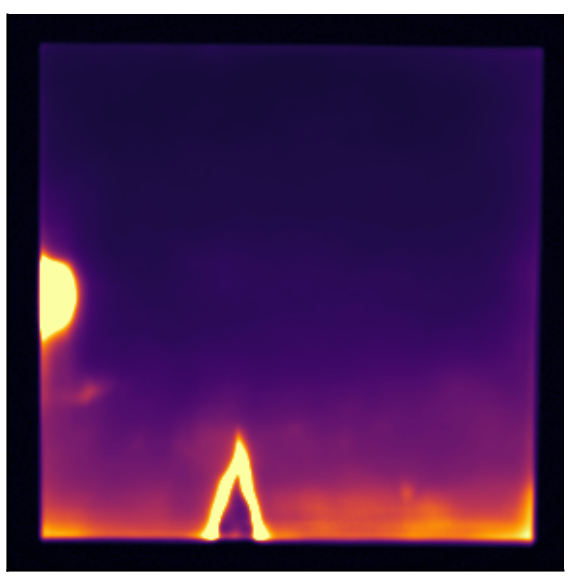


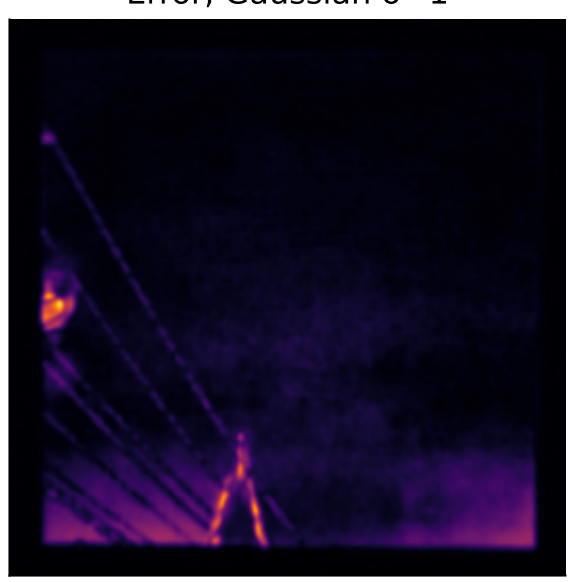


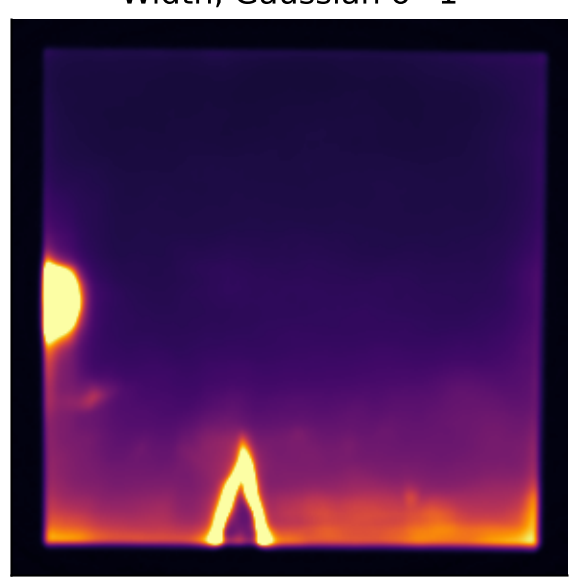


Ensemble (5)

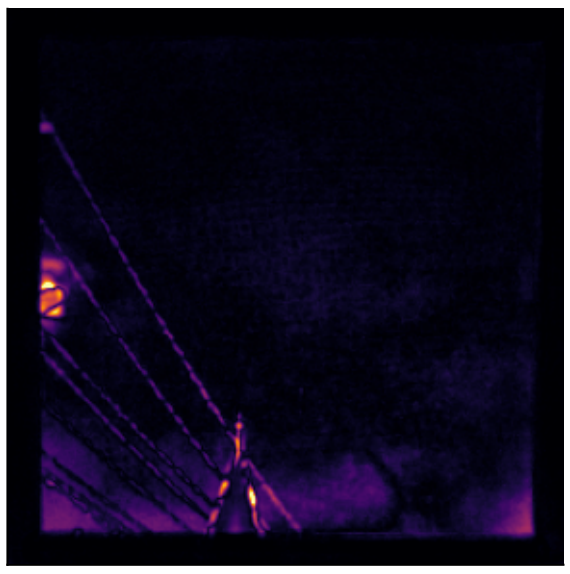


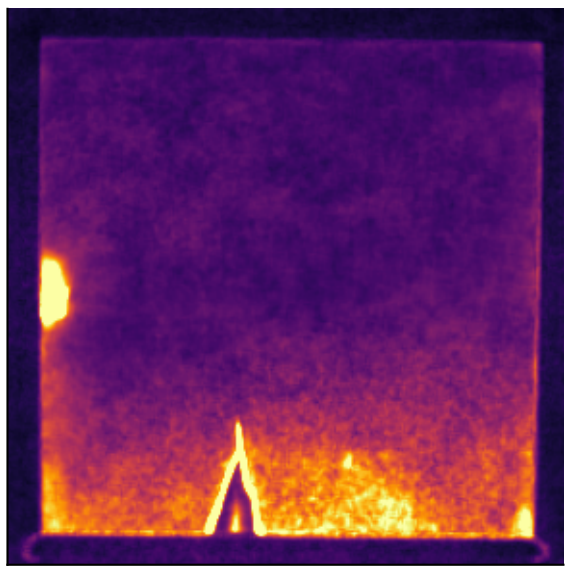


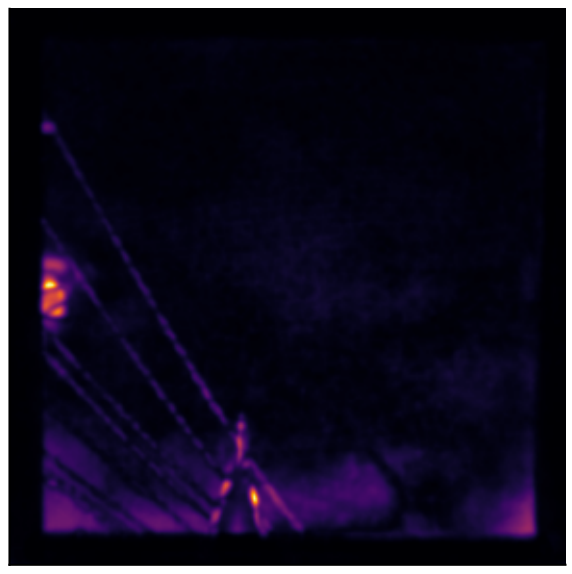


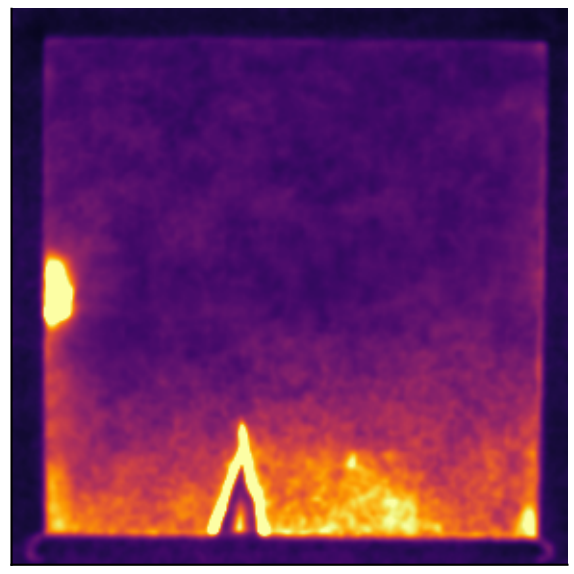


MC dropout (p=0.05, 30 passes)

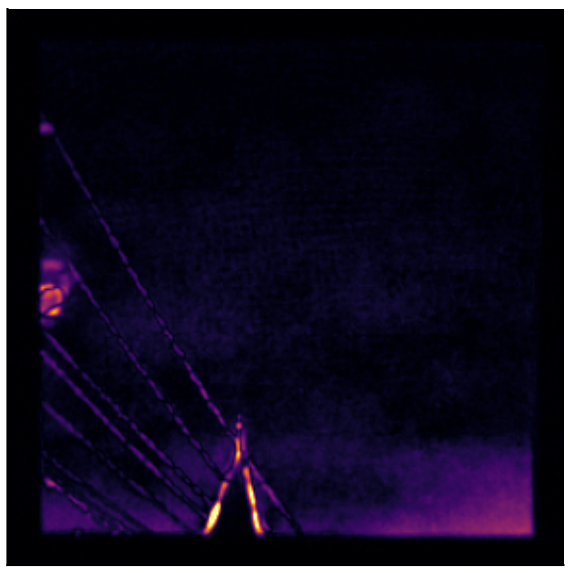


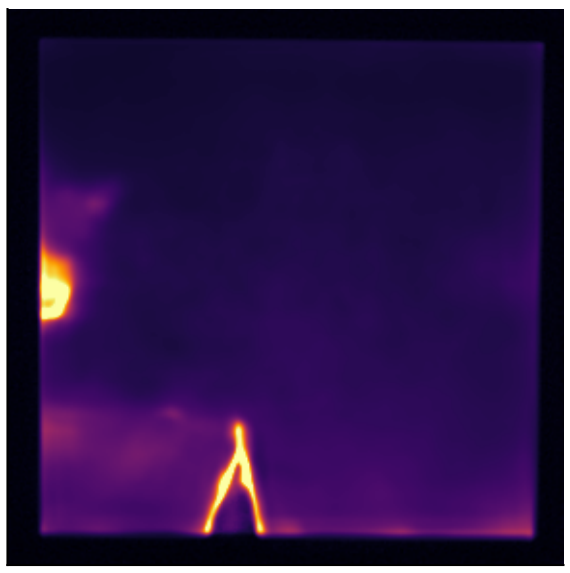


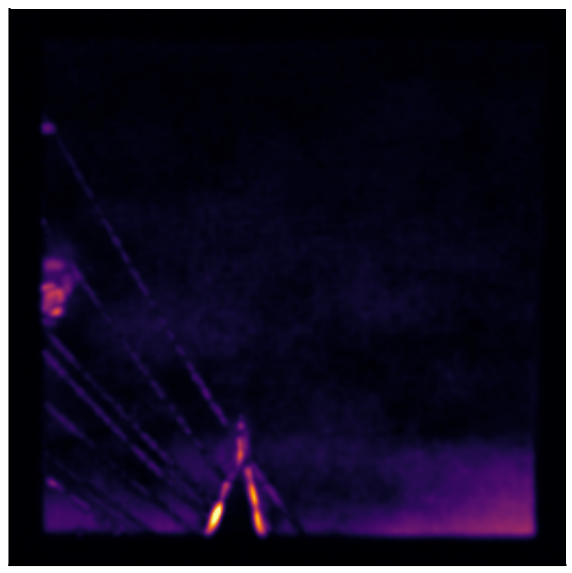


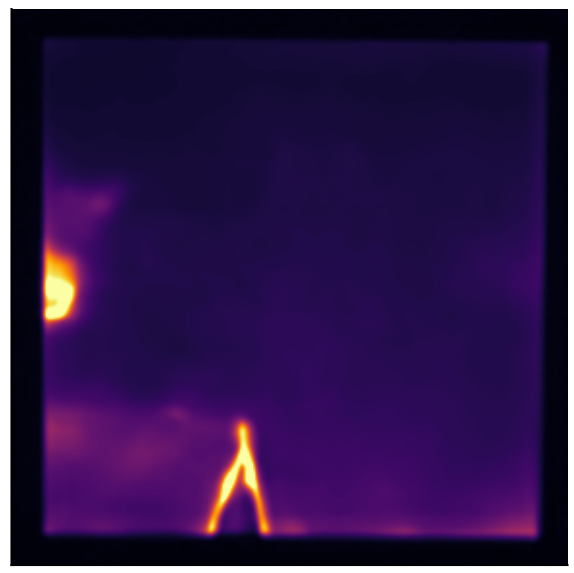


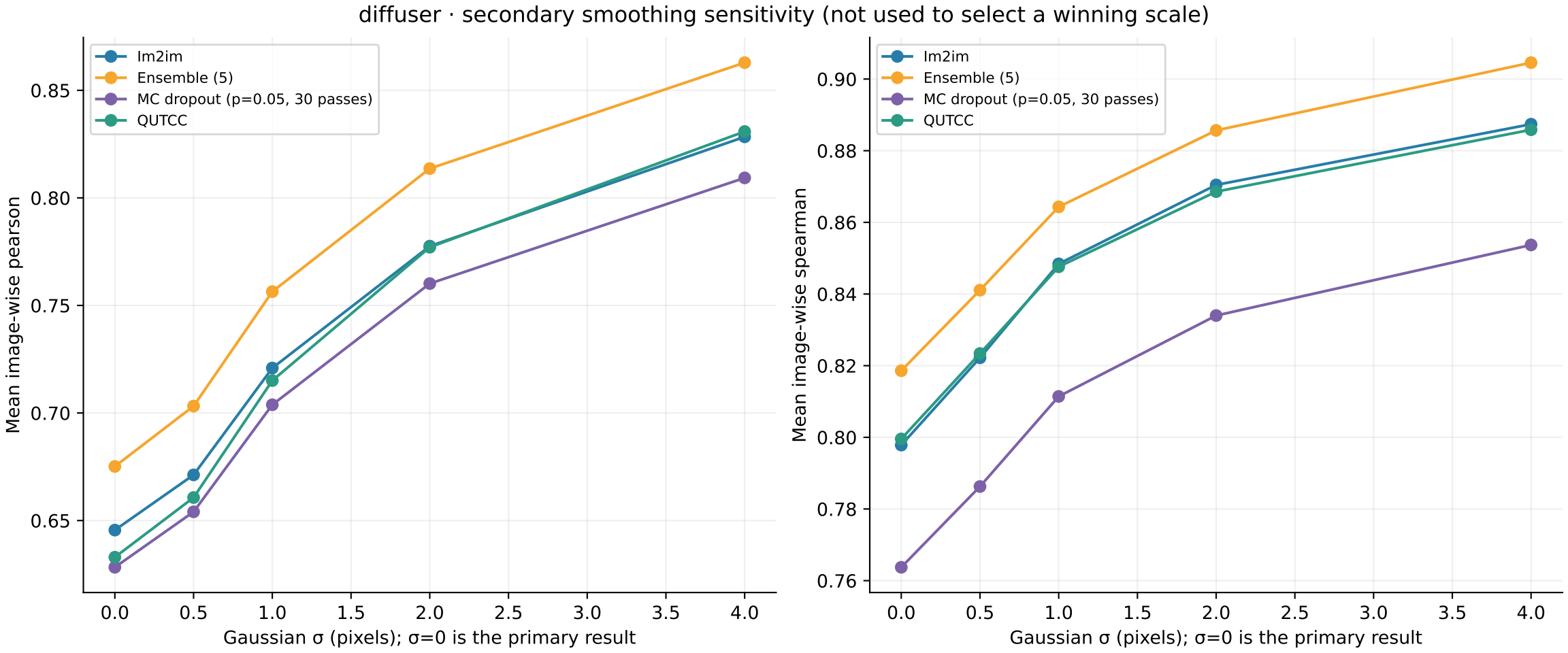
QUTCC










diffuser · secondary smoothing sensitivity (not used to select a winning scale)
Im2im
Ensemble (5)
MC dropout (p=0.05, 30 passes)
QUTCC
Mean image-wise pearson
0.85
0.80
0.75
0.70
0.65
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Gaussian σ (pixels); σ=0 is the primary result
Im2im
Ensemble (5)
MC dropout (p=0.05, 30 passes)
QUTCC
Mean image-wise spearman
0.90
0.88
0.86
0.84
0.82
0.80
0.78
0.76
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Gaussian σ (pixels); σ=0 is the primary result